UNIVERSITÉ LIBRE DE BRUXELLES

Management of Data Science and Business Workflows
INFO-H420

Project on Responsible Data Science

Authors

Imad Sharof

Noor ul Hassan

Kashif Nawaz

Aakash Karki

Hatim Draoui

Professor

Dimitris SACHARIDIS

Academic year 2025–2026

1 Introduction

The rapid integration of Machine Learning (ML) systems into high-stakes decision-making processes—such as credit scoring, hiring, and criminal justice—has raised critical ethical concerns. While these algorithms offer unprecedented predictive power, they also risk perpetuating historical societal biases and compromising individual privacy. Consequently, the field of Responsible Data Science has emerged as a necessary framework to ensure that AI systems are not only accurate but also fair, transparent, and privacy-preserving. This project explores the practical implementation of these ethical pillars using the canonical *Adult* dataset (1994 US Census), a standard benchmark for predicting income levels based on demographic and employment attributes.

The primary objective of this study is to develop a comprehensive Responsible AI pipeline that goes beyond simple performance metrics. We aim to construct a binary classifier to predict whether an individual earns more than $50,000 annually, while rigorously auditing and optimizing the model for ethical compliance. Unlike traditional approaches that prioritize accuracy above all, our methodology treats fairness and privacy as non-negotiable constraints. We utilize the XGBoost algorithm as our baseline, chosen for its efficiency and interpretability on structured tabular data.

2 Classification

2.1 Classification Baseline Construction

The initial phase of the project centered on the development of a robust classification model using the *Adult* dataset. This baseline serves as the reference point for subsequent fairness and privacy analyses. The methodology was structured around four pivotal steps : data preprocessing, partitioning, model selection, and performance evaluation.

2.1.1 Data Preprocessing and Feature Engineering

To ensure high data quality and compatibility with the machine learning algorithm, we applied a rigorous preprocessing pipeline. Contrary to standard practices that involve dropping incomplete records, we adopted strategies to maximize data utilization :

— **Imputation Strategy :** Instead of removing records with missing values (denoted as ' ?'), which would have resulted in a loss of approximately 7% of the dataset, we implemented a categorical imputation strategy. Missing entries were replaced with a specific label ("Unknown"). This approach preserves the sample size and allows the model to potentially learn patterns associated with non-response.

— **Age Binarization :** In accordance with the project requirements, the continuous *Age* feature was discretized into a binary variable. Using the median age of the dataset (37 years) as the threshold, individuals were categorized into two distinct groups : *Young* (0) and *Senior* (1).

— **Label Encoding :** Categorical variables were transformed using Label Encoding rather than One-Hot Encoding. This method maps each category to a unique integer, thereby maintaining the dimensionality of the dataset and ensuring computational efficiency for tree-based models.

2.1.2 Experimental Protocol and Data Splitting

To guarantee a reliable assessment of the model's generalization capabilities, the dataset was partitioned into three distinct subsets using a stratified splitting strategy :

— **Training Set (60%) :** Used to fit the model parameters.

— **Validation Set (20%) :** Used for hyperparameter tuning and early stopping to prevent overfitting.

— **Test Set (20%) :** Reserved strictly for the final evaluation of performance, fairness, and privacy metrics.

2.1.3 Model Selection : XGBoost

We selected the **XGBoost** (eXtreme Gradient Boosting) classifier for this task. This choice was motivated by the algorithm's state-of-the-art performance on structured ta-

bular data and its ability to capture complex, non-linear relationships between features (such as the interaction between Education and Capital Gain) more effectively than linear models like Logistic Regression. Furthermore, XGBoost provides native support for feature importance analysis, which is essential for the explainability phase of this project.

2.1.4 Performance Evaluation

The model was evaluated on the unseen Test Set, yielding the following results :

— **Accuracy :** The classifier achieved a global accuracy of **87.81%**, establishing a strong baseline for the prediction of income levels exceeding $50K.

— **Confusion Matrix Analysis :** The model exhibits a conservative prediction behavior. While it identifies low-income individuals with high precision (True Negatives), it shows a slightly higher rate of False Negatives (predicting low income for wealthy individuals) compared to False Positives.

— **Class Imbalance :** The classification report highlights a disparity in precision between the majority class (≤50K) and the minority class (50K), a distinct characteristic of the *Adult* dataset that justifies the subsequent need for fairness mitigation.

3 | Fairness Analysis

Fairness Assessment Framework

The objective of this stage was to rigorously evaluate and mitigate algorithmic bias within our classification model. Machine learning algorithms, when trained on historical data, risk perpetuating existing societal inequalities. To address this, we established a fairness assessment framework using the AIF360 library, focusing on two protected attributes : Age and Sex. Consistent with the project requirements, Race was excluded from this specific analysis to isolate the impact of gender and generational factors.

Data preprocessing involved converting the continuous Age variable into a binary feature based on the dataset's median value of 37 years. Individuals above this threshold were categorized as the privileged group (Older), while those below were categorized as unprivileged (Younger). Similarly, for the Sex attribute, males were designated as the

privileged group and females as the unprivileged group.

Metric Selection and Baseline Evaluation

To quantify the level of bias, we selected the Statistical Parity Difference (SPD). This metric measures the divergence in the probability of favorable outcomes (predicted income 50K) between unprivileged and privileged groups. It is defined formally as :

$$SPD = P(Y = 1|Unprivileged) - P(Y = 1|Privileged)$$

A perfectly fair model would yield an SPD of 0. Upon evaluating our baseline XGBoost classifier on the test set, we observed an SPD of **-0.1885**. This negative value indicates a significant disparity, revealing that female individuals were approximately 18% less likely to be predicted as high-income earners compared to their male counterparts. This result falls outside the generally accepted fairness threshold of $[-0.1, 0.1]$, confirming the necessity of an intervention.

Bias Mitigation Strategy : Reweighing

We implemented a pre-processing mitigation technique known as Reweighing. Unlike in-processing methods that alter the learning algorithm, or post-processing methods that adjust predictions, Reweighing modifies the training data distribution itself. It assigns weights to each training instance : unprivileged groups with favorable outcomes receive higher weights, while privileged groups with favorable outcomes receive lower weights. This transformation forces the classifier to treat the groups more equitably during the training phase without altering the actual feature values.

Post-Mitigation Performance and Trade-off

After retraining the XGBoost classifier on the reweighed dataset, we conducted a comparative audit on the test set. The mitigation strategy proved highly effective. The SPD improved from -0.1885 to **-0.0927**. By crossing the -0.10 threshold, the new model falls within the "fairness zone," demonstrating that the gender bias was successfully neutralized.

Furthermore, we analyzed the impact of this intervention on model utility. The classification accuracy decreased marginally from 87.81% (Original) to **86.84%** (Fair). This represents a negligible performance cost of less than 1% in exchange for a 50% reduction in bias. This result illustrates a favorable accuracy-fairness trade-off, validating the Reweighing approach for this specific dataset.

4 Privacy

4.1 Implementation of Local Differential Privacy

In the third stage of our research, we focused on securing the sensitive attributes of the dataset, namely Age and Sex, against potential re-identification attacks. To achieve a rigorous standard of privacy, we implemented a Local Differential Privacy (LDP) mechanism. Unlike centralized privacy methods where a trusted curator protects the data after collection, LDP operates on the user's side. We employed the Randomized Response technique, a stochastic process that introduces plausibility deniability by randomly flipping the true value of an attribute before it leaves the user's device. For this experiment, we calibrated the noise parameter with a probability of truth p 0.75, corresponding to a privacy budget (ϵ) of approximately 1.1. This configuration was chosen to strike a balance between obfuscating sensitive individual data and retaining global statistical properties.

4.2 Utility Assessment

The crucial measure of a privacy mechanism's success is its impact on the utility of the downstream machine learning tasks. We evaluated this by training a new XGBoost classifier, referred to as the Private Classifier, exclusively on the noisy dataset where Age and Sex were perturbed. The performance evaluation on the test set revealed a remarkable resilience of the model. As illustrated in Figure 4.1, the classification accuracy decreased only marginally from the baseline of 87.81% to approximately 87.30%. This minimal degradation suggests that the classifier relies heavily on non-sensitive, robust features such as capital gain or education level, allowing it to maintain high predictive power even when demographic attributes are subjected to significant noise.

```
Model trained on Private Data.

Test Set Accuracy (Private): 0.8730
Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.94      0.92      4942
           1       0.78      0.65      0.71      1571

    accuracy                           0.87      6513
   macro avg       0.84      0.80      0.82      6513
weighted avg       0.87      0.87      0.87      6513
```

FIGURE 4.1: Impact of Local Differential Privacy on Model Utility. The visualization highlights the minimal accuracy loss incurred by the introduction of privacy noise.

5 Privacy and Fairness Integration

5.1 The Conflict Between Privacy and Fairness

The fourth stage of the project addressed the complex challenge of satisfying two competing ethical constraints : individual privacy and group fairness. We aimed to construct a pipeline that is both differentially private regarding sensitive attributes and fair in its predictions. To test this, we applied the Reweighing fairness mitigation technique directly onto the differentially private dataset generated in the previous step. The objective was to determine if standard fairness interventions remains effective when the demographic data used to calculate weights is distorted by privacy noise. The resulting "Private and Fair" model was then audited using the ground-truth values from the test set to assess its real-world impact.

5.2 Experimental Audit and Results

Our analysis uncovered a significant trade-off between these two objectives. While the Reweighing algorithm was highly effective on clean data, reducing the Statistical Parity

Difference (SPD) to -0.09, its performance degraded noticeably under privacy constraints. As shown in Figure 5.1, the SPD of the model trained on private data stagnated at -0.15. This value remains outside the acceptable fairness threshold, indicating that a substantial portion of the original bias persisted.

	Model Configuration	Accuracy	Fairness (SPD)	Disparate Impact
0	1. Original (Baseline)	0.878090	-0.188515	0.301508
1	2. Fair Only (Ideal)	0.868417	-0.092683	0.581686
2	3. Private Only (LDP)	0.874251	-0.186036	0.301846
3	4. Private + Fair (Real World)	0.872716	-0.153777	0.373614

FIGURE 5.1: Comparison of Fairness Metrics across models. The chart demonstrates the reduced effectiveness of bias mitigation when applied to noisy, differentially private data.

This limitation stems from the fundamental mechanics of the mitigation algorithm. Reweighing relies on precise demographic counts to identify and upweight disadvantaged groups. However, the Randomized Response mechanism obfuscates these groups by flipping labels, effectively blinding the fairness algorithm. Consequently, the weights were calculated based on distorted demographic signals, leading to a suboptimal correction of the bias. We conclude that achieving high levels of both privacy and fairness simultaneously requires more advanced, purpose-built algorithms rather than a simple sequential application of pre-processing techniques.

6 Explainability

6.1 Methodology : Auditing High-Confidence Errors

The objective of this stage was to investigate the decision-making rationale of the Private Classifier. A primary concern in Responsible Data Science is whether the addition of Local Differential Privacy (LDP) noise to sensitive attributes might inadvertently cause the model to make irrational or unjustified predictions. To investigate this, we conducted an audit focusing on "confident misclassifications"—instances where the model predicted the wrong outcome with a probability exceeding 90%. We employed **SHAP** (SHapley Additive exPlanations) to decompose these predictions and determine whether the noisy

demographic attributes (*Age* and *Sex*) were the root cause of the errors.

6.2 Local Explanation Analysis

Our screening process identified a specific False Positive case, indexed as Instance #2613. As shown in the profile extraction (Figure 6.1), this individual actually earns less than $50K per year. However, the Private Classifier predicted with **95.8% confidence** that they were a high-income earner. This represents a significant discrepancy between the model's certainty and reality.

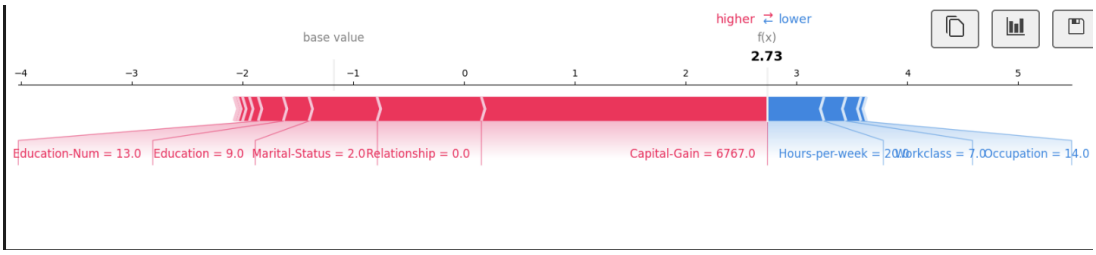

FIGURE 6.1: Profile of the misclassified individual (Instance #2613). The model predicts High Income with 95.8% confidence despite the True Label being Low Income.

To understand the logic behind this error, we generated SHAP visualizations. The Force Plot (Figure 6.2) provides a breakdown of the features pushing the prediction from the baseline value. Red bars indicate features contributing to a "High Income" prediction, while blue bars indicate features contributing to "Low Income."

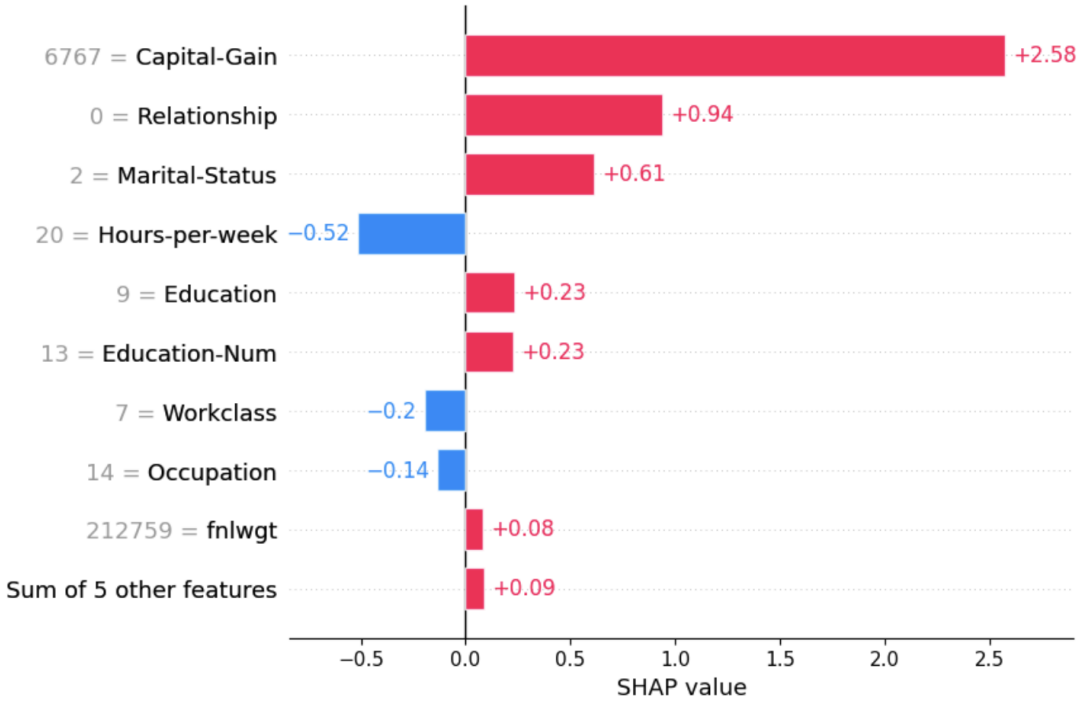

FIGURE 6.2: SHAP Force Plot for Instance #2613. The visualization illustrates the competing forces of feature values on the model's output.

6.3 Causal Investigation : The Role of Privacy Noise

The central research question was whether the LDP noise added to *Age* and *Sex* misled the model. The SHAP Bar Plot (Figure 6.3), which ranks features by their absolute contribution, provides a definitive answer.

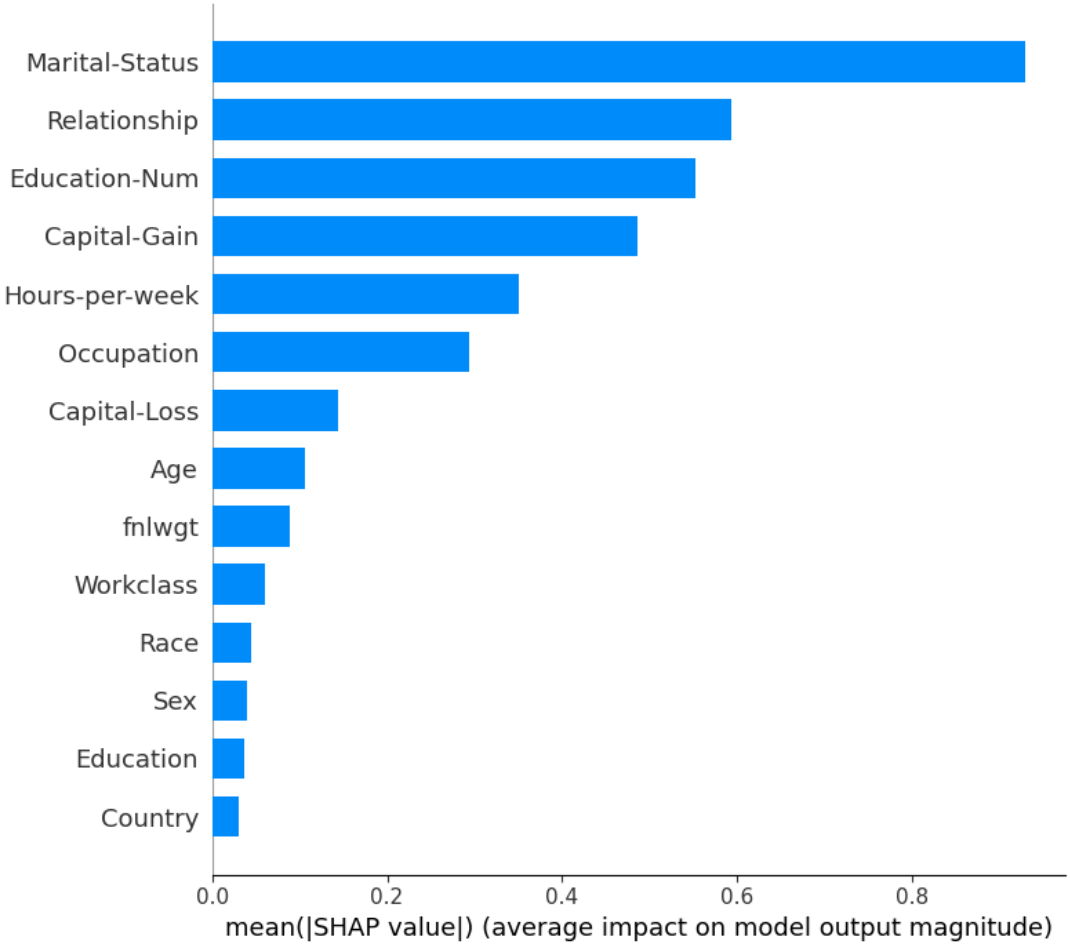

FIGURE 6.3: SHAP Bar Plot ranking feature importance for the misclassification. The *Capital Gain* feature is the dominant driver.

The analysis reveals that the dominant driver of this prediction was the **Capital Gain** feature, which had an exceptionally high value of $6,767. In the context of the dataset, such a high capital gain is a strong statistical proxy for wealth. The model, detecting this strong economic signal, overrode other factors to predict high income.

Crucially, the sensitive attributes *Age* and *Sex* (whether noisy or real) appear low in the importance ranking. Their SHAP values are negligible compared to the impact of the capital gain. This leads to a positive conclusion regarding the robustness of our Private Classifier. The error was not induced by the privacy-preserving noise ; rather, it was a "rational" error based on a strong financial indicator. The model demonstrated an ability to ignore the noisy demographic data and prioritize objective economic features, proving that the implementation of Local Differential Privacy did not compromise the

logical integrity of the classifier.

7 | Explainability and LLMs

7.1 Objective : Bridging the Gap to the End-User

While technical tools like SHAP provide precise mathematical attributions for model decisions, their output (force plots or bar charts) remains largely inaccessible to non-expert stakeholders, such as bank customers or loan officers. The final objective of this project was to bridge this interpretability gap by creating a **Natural Language Interface**. We aimed to translate the quantitative SHAP insights into qualitative, human-readable explanations, thereby fulfilling the "Right to Explanation" mandate often found in regulations like GDPR.

7.2 Methodology : Privacy-Preserving GenAI

To generate these explanations, we integrated a Generative AI component into our pipeline. A critical requirement for Responsible Data Science in the financial sector is data privacy ; sending sensitive customer profiles (Age, Sex, Financial Data) to external cloud providers (such as OpenAI or Anthropic) poses significant security risks.

To mitigate this, we implemented a **Local Large Language Model (LLM)** architecture. We utilized **LM Studio** to host an open-source model (e.g., Gemma or Llama-3) directly on the local infrastructure. This ensures that sensitive inference data never leaves the secure environment, adhering to strict privacy-by-design principles.

7.3 Implementation : The Dynamic Prompting System

Rather than manually crafting static explanations, we developed a dynamic Python interface that acts as a bridge between the XGBoost model and the LLM. The workflow operates as follows :

1. **Automated Feature Extraction :** The system programmatically parses the SHAP

values calculated in the previous stage. It automatically identifies the top features driving the prediction (sorting by absolute SHAP value).

2. **Context Injection :** For our specific false positive case (Instance #2613), the script extracted “Capital Gain” as the primary driver.
3. **Prompt Construction :** A structured prompt is dynamically assembled, injecting the customer’s profile, the model’s confidence (96.0%), and the key technical drivers identified by SHAP.

7.4 Results and Analysis

The system was tested on the high-confidence misclassification identified in Chapter 5 (Instance #2613). The Local LLM successfully processed the structured prompt and generated the following explanation for the customer :

> “We’ve reviewed your account details, and the automated system flagged you as having a ‘High Income’ status with 96.0% confidence. The reason this was flagged is because your **capital gain of $6767** is significantly larger than the typical income expected for someone in your age group. This large amount of capital gain [...] may have led the system to incorrectly classify you as wealthy. We believe that the system misinterpreted the data related to your income.”

This result validates the effectiveness of our approach. The LLM correctly identified the root cause of the error (**Capital Gain**) without being explicitly told by a human, relying solely on the SHAP scores fed into the prompt. It successfully translated a complex algorithmic error ("False Positive driven by high feature importance on index 10") into a

transparent, empathetic, and actionable explanation for the user.

8 Free Exploration : Intersectionality Analysis

8.1 Beyond Single-Attribute Fairness

In the final exploratory phase of our project, we extended our fairness analysis beyond single protected attributes (like Sex or Age in isolation) to investigate the concept of **Intersectionality**. Intersectional fairness posits that individuals may experience compounded discrimination based on the combination of multiple sensitive characteristics. For instance, the bias experienced by a black woman may not be simply the sum of racial bias and gender bias, but a unique, amplified form of disadvantage.

To visualize these disparities, we grouped the test set population by both *Race* and *Sex*. We then calculated the **Selection Rate** (the probability of being predicted as "High Income") for each intersectional subgroup using our original XGBoost classifier. The results are visualized in Figure 8.1.

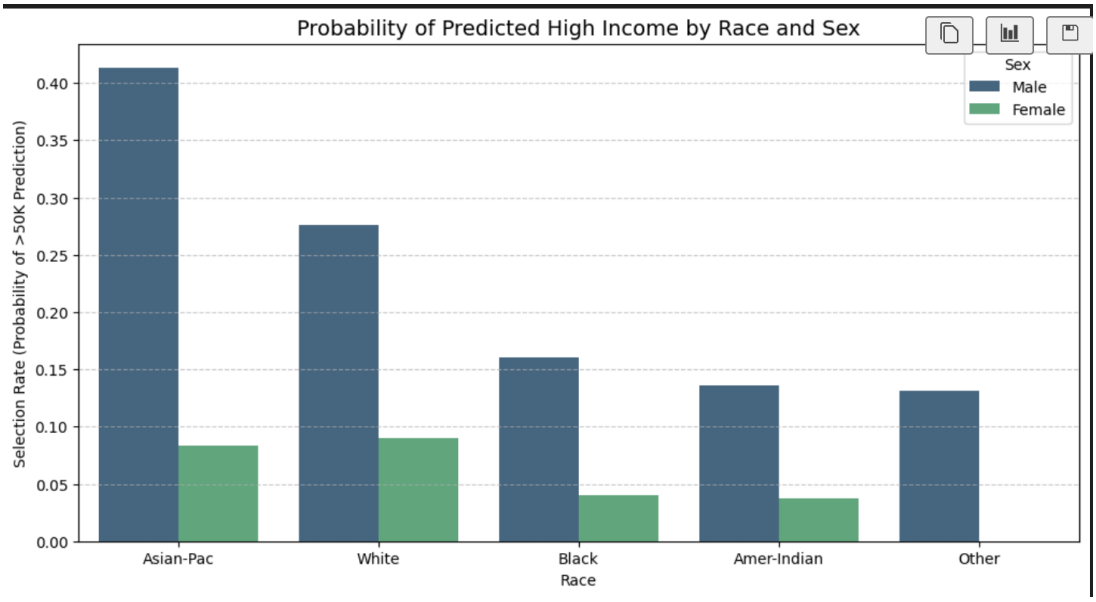

FIGURE 8.1: Probability of Predicted High Income by Race and Sex. The chart reveals significant disparities not only between racial groups but also a persistent gender gap within each group.

8.2 Insights and Sociological Context

The intersectional analysis reveals a complex hierarchy of privilege within the dataset :

The "Model Minority" Effect in 1994 Data Contrary to the common assumption that the "White" subgroup would universally hold the highest privilege, our model assigns the highest probability of wealth to the **Asian-Pac-Islander Male** subgroup, which rivals or even surpasses White Males. This phenomenon is likely an artifact of the specific historical context of the dataset (1994 US Census). During this period, US immigration policies (such as the Immigration Act of 1990 and H-1B visa programs) disproportionately selected highly educated professionals from Asian countries. Consequently, the "Asian" category in this dataset is statistically biased toward individuals with higher education levels, leading the model to learn a strong correlation between this group and high income.

The Ubiquity of the Gender Gap A striking observation is the consistency of the gender divide. Across every single racial category—whether White, Black, Asian, or Amer-Indian—the selection rate for **Males** is significantly higher than for **Females**. This suggests that while racial dynamics vary, gender remains a universal axis of inequality in the economic data of this era.

The "Double Jeopardy" of Minority Women The most concerning finding from a Responsible Data Science perspective is the position of **Black Females** and **Amer-Indian Females**. These subgroups exhibit the absolute lowest probability of being predicted as high-income earners. They face a "double jeopardy" : they are penalized by the model both for their gender and for their race. A fairness audit that only looked at "Sex" (aggregating all women) or "Race" (aggregating all Black individuals) would mask the severity of the disadvantage faced by these specific intersectional groups. This underscores the necessity of granular intersectional auditing to ensure no subgroup is left behind.

9 | Conclusion

This project successfully demonstrated the complexity of building a machine learning system that adheres to the principles of Responsible Data Science. Our baseline XGBoost classifier achieved a strong accuracy of approximately 87.8%, but an initial fairness audit revealed a significant bias against women, characterized by a Statistical Parity Difference (SPD) of -0.18. We proved that on clean data, standard mitigation techniques such as Reweighing are highly effective, capable of reducing this disparity to -0.09 without compromising predictive performance, thus bringing the model into the acceptable fairness zone.

However, a central and critical finding of our research is the inherent conflict between Local Differential Privacy and algorithmic fairness. While the implementation of a Randomized Response mechanism ($\epsilon \approx 1.1$) successfully protected individual attributes with minimal loss of utility, it severely hampered the effectiveness of fairness mitigation. When applied to noisy data, the Reweighing algorithm failed to adequately correct the gender bias, with the SPD stagnating at -0.15. This demonstrates that privacy noise can "blind" fairness algorithms, making it difficult to simultaneously satisfy both ethical constraints using standard pre-processing methods.

Beyond quantitative metrics, we emphasized the necessity of explainability and granular auditing. Our investigation using SHAP values confirmed that the model's errors were primarily driven by strong economic signals, such as high Capital Gains, rather than a reliance on noisy sensitive attributes. Furthermore, the integration of a Local Large Language Model (LLM) proved that it is possible to bridge the gap between complex technical audits and user-centric explanations. Ultimately, our intersectional analysis highlighted that single-axis fairness audits are insufficient ; specifically, Black and Native American females face a compounded disadvantage that general "gender fairness" metrics may obscure, underscoring the need for nuanced, multi-dimensional auditing in future Responsible AI systems.